Supplementary material

Does seismic-aware form-finding lower the embodied carbon of reinforced concrete shells? A pre-registered comparison of 168 sized designs

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S1 Literature search protocol

The gap statement of Section 1 rests on a documented search, run on 13 November 2026 in OpenAlex (about 250 million records, open access and paywalled). The search window was 2015–2026, to include the work of Michiels and Adriaenssens (2017–2019); the gap claim rests on papers from 2020–2026. A paper was included if it studies (a) form-finding or shape design of shells or arches under lateral or seismic load, (b) the embodied carbon of concrete shells or shell floors, or (c) the seismic response of concrete shells. Retrofit, frames, walls, tanks, piles and masonry infill were excluded.

Table S1: Search queries, screening and inclusion.

Query	Words	From	Scre
Q1	form finding shell structures seismic lateral loading	2015	t
Q2	embodied carbon concrete shell structures form-finding	2018	t
Q3	thin concrete shell roof seismic behaviour reinforced concrete earthquake	2015	t
Q4	earthquake response of thin reinforced concrete shell roofs stiffness low mass parametric	2016	t
Q5	concrete shell form finding optimisation material reduction buckling governs thickness	2018	t
Q6	embodied carbon seismic design shell form-found compression-only lateral load material efficiency	2019	t

Q6 found no paper that reports the embodied carbon of seismic-aware form-finding. Forward citation searching from Michiels and Adriaenssens (2017, 2018) is still to be done[check]. The full log, with the 15 included papers, is `paper/13_literature_search.md` in the repository.

S2 Verification of the analysis chain

S3 Pre-registration amendments

The pre-registration of 28–30 October 2026 and every dated amendment are in `docs/preregistration.md`. Each amendment states whether it was written before or after the results it affects. In summary:

Date (2026)	Timing	Change
30 Oct	before any run	Seismic-aware strategy defined (four directions, envelope check); own 3D solver; ICE replaced by a ±25 % factor band; thresholds set (λ 5, with 3 and 8; $L/300$; 50 mm; 5 mm steps; 5 % decision band). Midrange instead of mean of the four thrust surfaces.
31 Oct	before any design-space run	Corner-supported free edges drawn in by $0.10L$ (D3-1); own S8R deck, to be cross-checked against ShellBench's wrapper.

Date (2026)	Timing	Change
1 Nov	before the first sizing run	Seismic case in $+x$ only (symmetry); corner zones of $0.2L$ out of the compression check; characteristic deflection combination; bisection sizing.
1 Nov	after the 84 designs	Poisson's ratio read in place from ShellBench (value unchanged); loads per plan area recorded (D4-5); code basis for roof live load added. No thickness changed.
2 Nov	before the 84-design run	Steel rule (two directions, EN 1992-1-1 Annex F, no biaxial relief), minimum ratio in both directions, support-zone thickening, net steel.
3 Nov	after results; fix before the 20 m run	Carbon per m^2 of plan area for corner-supported shells (code followed the stated rule).
4 Nov	just after the S4 run	S4 fallback: closed-form lean of a parabolic arch (published arch not available).
6 Nov	during the sensitivity runs	How the three sensitivity runs were run (all 168 designs).
7 Nov	before the corrected runs	Pin moved to ShellBench v0.2.0 (0 mismatches); four buckling factors requested (D10-1).
8 Nov	citation only	EN 1992-1-1 Annex F read in full.
9 Nov	before the freeze run	Freeze one day early; pin ShellBench v0.3.0.
9 Nov	after the freeze	Presentation only (plotting style).
11 Nov	citation only	Buckling-threshold sources read (Dulácska and Kollár 1995; ACI 318.2-14).
14 Nov	after the freeze, exploratory	Verification checks V1–V5 and Fig. 6; no result changed.

S4 All 168 sized designs

Table S1 lists every sized design at the central carbon factors, with the low and high ends of the ±25 % band. Design identifiers give span, supports, rise-to-span ratio, strategy and S_a (g). The same values are in `paper/tables/tableS1_all_designs.csv`; the full record, with every check, is `results/batch_all.csv` at tag `freeze-2026-11-09` (ShellBench v0.3.0).

Table S3: All 168 sized designs.

design	t [mm]	governs	λ [-]	T1 [s]	steel [kg]	carbon [kgCO_2e/m^2]	band low	band high
L10-corners-r0.10-gravity-sa0.0	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.1	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.2	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.3	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.4	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.5	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-gravity-sa0.6	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.0	105	buckling	5.05	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.1	105	buckling	5.06	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.2	105	buckling	5.07	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.3	105	buckling	5.08	0.184	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.4	105	buckling	5.09	0.183	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.5	105	buckling	5.10	0.183	275	45.5	34.1	56.9
L10-corners-r0.10-seismic-sa0.6	105	buckling	5.10	0.182	275	45.5	34.1	56.9
L10-corners-r0.15-gravity-sa0.0	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-gravity-sa0.1	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-gravity-sa0.2	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-gravity-sa0.3	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-gravity-sa0.4	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-gravity-sa0.5	90	buckling	5.60	0.159	238	39.8	29.9	49.8

design	t [mm]	governs	λ [-]	T1 [s]	steel [kg]	carbon [kgCO_2e/m^2]	band low	band high
L10-corners-r0.15-gravity-sa0.6	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-seismic-sa0.0	90	buckling	5.60	0.159	238	39.8	29.9	49.8
L10-corners-r0.15-seismic-sa0.1	85	buckling	5.02	0.165	225	37.7	28.3	47.1
L10-corners-r0.15-seismic-sa0.2	85	buckling	5.04	0.164	225	37.7	28.2	47.1
L10-corners-r0.15-seismic-sa0.3	85	buckling	5.05	0.164	225	37.6	28.2	47.1
L10-corners-r0.15-seismic-sa0.4	85	buckling	5.05	0.165	225	37.6	28.2	47.0
L10-corners-r0.15-seismic-sa0.5	85	buckling	5.03	0.167	224	37.6	28.2	47.0
L10-corners-r0.15-seismic-sa0.6	90	buckling	5.63	0.161	237	39.7	29.8	49.7
L10-corners-r0.20-gravity-sa0.0	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.1	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.2	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.3	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.4	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.5	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-gravity-sa0.6	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-seismic-sa0.0	75	buckling	5.41	0.165	204	34.3	25.7	42.9
L10-corners-r0.20-seismic-sa0.1	75	buckling	5.50	0.164	204	34.3	25.7	42.9
L10-corners-r0.20-seismic-sa0.2	75	buckling	5.54	0.164	204	34.3	25.7	42.9
L10-corners-r0.20-seismic-sa0.3	75	buckling	5.52	0.167	203	34.3	25.7	42.8
L10-corners-r0.20-seismic-sa0.4	75	buckling	5.45	0.170	203	34.2	25.6	42.7
L10-corners-r0.20-seismic-sa0.5	75	buckling	5.34	0.175	202	34.1	25.6	42.6
L10-corners-r0.20-seismic-sa0.6	75	buckling	5.26	0.180	203	34.0	25.5	42.5
L10-edges-r0.10-gravity-sa0.0	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.1	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.2	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.3	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.4	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.5	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-gravity-sa0.6	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.0	50	$t_{\min}$	67.93	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.1	50	$t_{\min}$	68.09	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.2	50	$t_{\min}$	68.79	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.3	50	$t_{\min}$	70.13	0.045	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.4	50	$t_{\min}$	72.38	0.044	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.5	50	$t_{\min}$	75.25	0.044	173	21.6	16.2	27.0
L10-edges-r0.10-seismic-sa0.6	50	$t_{\min}$	78.51	0.044	173	21.6	16.2	27.0
L10-edges-r0.15-gravity-sa0.0	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.1	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.2	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.3	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.4	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.5	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-gravity-sa0.6	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.0	50	$t_{\min}$	114.94	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.1	50	$t_{\min}$	115.78	0.037	179	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.2	50	$t_{\min}$	119.35	0.036	179	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.3	50	$t_{\min}$	126.66	0.036	180	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.4	50	$t_{\min}$	136.84	0.035	180	22.4	16.8	28.0
L10-edges-r0.15-seismic-sa0.5	50	$t_{\min}$	149.31	0.034	180	22.4	16.8	28.1
L10-edges-r0.15-seismic-sa0.6	50	$t_{\min}$	163.12	0.034	180	22.5	16.9	28.1
L10-edges-r0.20-gravity-sa0.0	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.1	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.2	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.3	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.4	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.5	50	$t_{\min}$	146.75	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-gravity-sa0.6	50	$t_{\min}$	140.91	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-seismic-sa0.0	50	$t_{\min}$	149.42	0.034	188	23.4	17.6	29.3
L10-edges-r0.20-seismic-sa0.1	50	$t_{\min}$	152.00	0.033	188	23.4	17.6	29.3
L10-edges-r0.20-seismic-sa0.2	50	$t_{\min}$	161.99	0.033	188	23.4	17.6	29.3
L10-edges-r0.20-seismic-sa0.3	50	$t_{\min}$	181.70	0.032	188	23.5	17.6	29.4
L10-edges-r0.20-seismic-sa0.4	50	$t_{\min}$	210.58	0.031	189	23.6	17.7	29.4
L10-edges-r0.20-seismic-sa0.5	50	$t_{\min}$	239.06	0.029	190	23.6	17.7	29.5
L10-edges-r0.20-seismic-sa0.6	50	$t_{\min}$	252.17	0.029	190	23.8	17.8	29.7
L20-corners-r0.10-gravity-sa0.0	265	buckling	5.17	0.307	2895	116.9	87.7	146.1
L20-corners-r0.10-gravity-sa0.1	265	buckling	5.17	0.307	2895	116.9	87.7	146.1
L20-corners-r0.10-gravity-sa0.2	265	buckling	5.17	0.307	2897	116.9	87.7	146.1
L20-corners-r0.10-gravity-sa0.3	265	buckling	5.17	0.307	2901	116.9	87.7	146.2
L20-corners-r0.10-gravity-sa0.4	265	buckling	5.17	0.307	2907	117.0	87.7	146.2
L20-corners-r0.10-gravity-sa0.5	265	buckling	5.17	0.307	2929	117.2	87.9	146.4
L20-corners-r0.10-gravity-sa0.6	265	buckling	5.17	0.307	2953	117.3	88.0	146.7
L20-corners-r0.10-seismic-sa0.0	265	buckling	5.17	0.307	2895	116.9	87.7	146.1
L20-corners-r0.10-seismic-sa0.1	265	buckling	5.18	0.307	2895	116.9	87.7	146.1

design	t [mm]	governs	λ [-]	T1 [s]	steel [kg]	carbon [kgCO_2e/m^2]	band low	band high
L20-corners-r0.10-seismic-sa0.2	265	buckling	5.18	0.307	2896	116.9	87.7	146.1
L20-corners-r0.10-seismic-sa0.3	265	buckling	5.19	0.306	2898	116.9	87.7	146.2
L20-corners-r0.10-seismic-sa0.4	265	buckling	5.20	0.306	2902	116.9	87.7	146.2
L20-corners-r0.10-seismic-sa0.5	260	buckling	5.00	0.308	2864	114.9	86.2	143.6
L20-corners-r0.10-seismic-sa0.6	260	buckling	5.01	0.308	2882	115.0	86.3	143.8
L20-corners-r0.15-gravity-sa0.0	215	buckling	5.04	0.261	2321	96.1	72.0	120.1
L20-corners-r0.15-gravity-sa0.1	215	buckling	5.04	0.261	2321	96.1	72.0	120.1
L20-corners-r0.15-gravity-sa0.2	215	buckling	5.04	0.261	2321	96.1	72.0	120.1
L20-corners-r0.15-gravity-sa0.3	215	buckling	5.04	0.261	2321	96.1	72.0	120.1
L20-corners-r0.15-gravity-sa0.4	215	buckling	5.04	0.261	2322	96.1	72.0	120.1
L20-corners-r0.15-gravity-sa0.5	215	buckling	5.04	0.261	2324	96.1	72.1	120.1
L20-corners-r0.15-gravity-sa0.6	215	buckling	5.04	0.261	2336	96.2	72.1	120.2
L20-corners-r0.15-seismic-sa0.0	215	buckling	5.04	0.261	2321	96.1	72.0	120.1
L20-corners-r0.15-seismic-sa0.1	215	buckling	5.06	0.261	2321	96.1	72.1	120.1
L20-corners-r0.15-seismic-sa0.2	215	buckling	5.08	0.259	2321	96.1	72.1	120.1
L20-corners-r0.15-seismic-sa0.3	215	buckling	5.09	0.257	2320	96.0	72.0	120.0
L20-corners-r0.15-seismic-sa0.4	215	buckling	5.09	0.253	2318	96.0	72.0	120.0
L20-corners-r0.15-seismic-sa0.5	215	buckling	5.09	0.257	2319	95.9	71.9	119.9
L20-corners-r0.15-seismic-sa0.6	215	buckling	5.08	0.257	2335	95.9	72.0	119.9
L20-corners-r0.20-gravity-sa0.0	185	buckling	5.14	0.260	2035	85.2	63.9	106.4
L20-corners-r0.20-gravity-sa0.1	185	buckling	5.14	0.260	2035	85.2	63.9	106.4
L20-corners-r0.20-gravity-sa0.2	185	buckling	5.14	0.260	2035	85.2	63.9	106.4
L20-corners-r0.20-gravity-sa0.3	185	buckling	5.14	0.260	2035	85.2	63.9	106.4
L20-corners-r0.20-gravity-sa0.4	185	buckling	5.14	0.260	2036	85.2	63.9	106.5
L20-corners-r0.20-gravity-sa0.5	185	buckling	5.14	0.260	2039	85.2	63.9	106.5
L20-corners-r0.20-gravity-sa0.6	185	buckling	5.14	0.260	2047	85.2	63.9	106.6
L20-corners-r0.20-seismic-sa0.0	185	buckling	5.14	0.260	2035	85.2	63.9	106.4
L20-corners-r0.20-seismic-sa0.1	185	buckling	5.20	0.260	2035	85.2	63.9	106.5
L20-corners-r0.20-seismic-sa0.2	185	buckling	5.23	0.259	2034	85.1	63.8	106.4
L20-corners-r0.20-seismic-sa0.3	185	buckling	5.22	0.262	2030	85.0	63.8	106.3
L20-corners-r0.20-seismic-sa0.4	185	buckling	5.18	0.267	2028	84.9	63.6	106.1
L20-corners-r0.20-seismic-sa0.5	185	buckling	5.11	0.273	2033	84.7	63.5	105.9
L20-corners-r0.20-seismic-sa0.6	185	buckling	5.08	0.279	2057	84.7	63.5	105.8
L20-edges-r0.10-gravity-sa0.0	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.1	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.2	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.3	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.4	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.5	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-gravity-sa0.6	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.0	50	$t_{\min}$	14.00	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.1	50	$t_{\min}$	14.04	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.2	50	$t_{\min}$	14.07	0.111	692	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.3	50	$t_{\min}$	14.32	0.110	693	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.4	50	$t_{\min}$	14.84	0.109	693	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.5	50	$t_{\min}$	15.54	0.108	693	21.6	16.2	27.0
L20-edges-r0.10-seismic-sa0.6	50	$t_{\min}$	16.32	0.107	693	21.6	16.2	27.0
L20-edges-r0.15-gravity-sa0.0	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.1	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.2	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.3	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.4	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.5	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-gravity-sa0.6	50	$t_{\min}$	23.14	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.0	50	$t_{\min}$	23.43	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.1	50	$t_{\min}$	23.65	0.090	718	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.2	50	$t_{\min}$	24.50	0.089	718	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.3	50	$t_{\min}$	26.10	0.087	718	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.4	50	$t_{\min}$	28.76	0.086	719	22.4	16.8	28.0
L20-edges-r0.15-seismic-sa0.5	50	$t_{\min}$	31.27	0.084	720	22.4	16.8	28.1
L20-edges-r0.15-seismic-sa0.6	50	$t_{\min}$	33.50	0.082	722	22.5	16.9	28.1
L20-edges-r0.20-gravity-sa0.0	50	$t_{\min}$	30.31	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.1	50	$t_{\min}$	30.31	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.2	50	$t_{\min}$	30.31	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.3	50	$t_{\min}$	30.31	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.4	50	$t_{\min}$	29.87	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.5	50	$t_{\min}$	28.49	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-gravity-sa0.6	50	$t_{\min}$	27.22	0.081	752	23.4	17.6	29.3
L20-edges-r0.20-seismic-sa0.0	50	$t_{\min}$	30.31	0.081	751	23.4	17.6	29.3
L20-edges-r0.20-seismic-sa0.1	50	$t_{\min}$	30.95	0.080	751	23.4	17.6	29.3
L20-edges-r0.20-seismic-sa0.2	50	$t_{\min}$	33.35	0.079	752	23.4	17.6	29.3
L20-edges-r0.20-seismic-sa0.3	50	$t_{\min}$	38.07	0.076	753	23.5	17.6	29.4
L20-edges-r0.20-seismic-sa0.4	50	$t_{\min}$	44.24	0.074	755	23.6	17.7	29.4

design	t [mm]	governs	λ [-]	T1 [s]	steel [kg]	carbon [kgCO_2e/m^2]	band low	band high
L20-edges-r0.20-seismic-sa0.5	50	$t_{\min}$	49.13	0.071	758	23.6	17.7	29.5
L20-edges-r0.20-seismic-sa0.6	50	$t_{\min}$	52.89	0.069	762	23.8	17.8	29.7

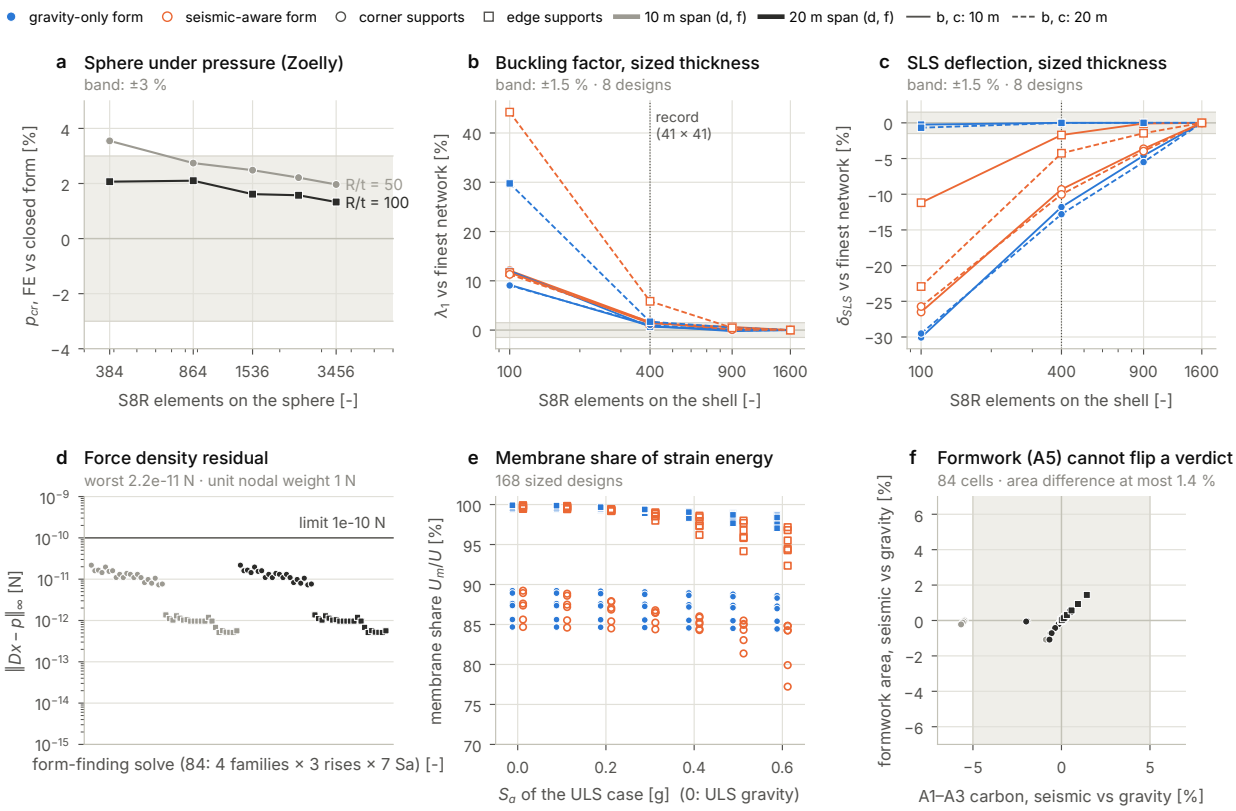


Figure S1: Verification checks V1–V5, run after the freeze (Section 3.6 of the paper): (a) sphere buckling against the classical value; (b) first buckling factor and (c) SLS deflection under mesh refinement from 21 to 81 nodes per side; (d) residual of the assembled force density system; (e) membrane share of strain energy; (f) soffit area difference against carbon difference (formwork bound).